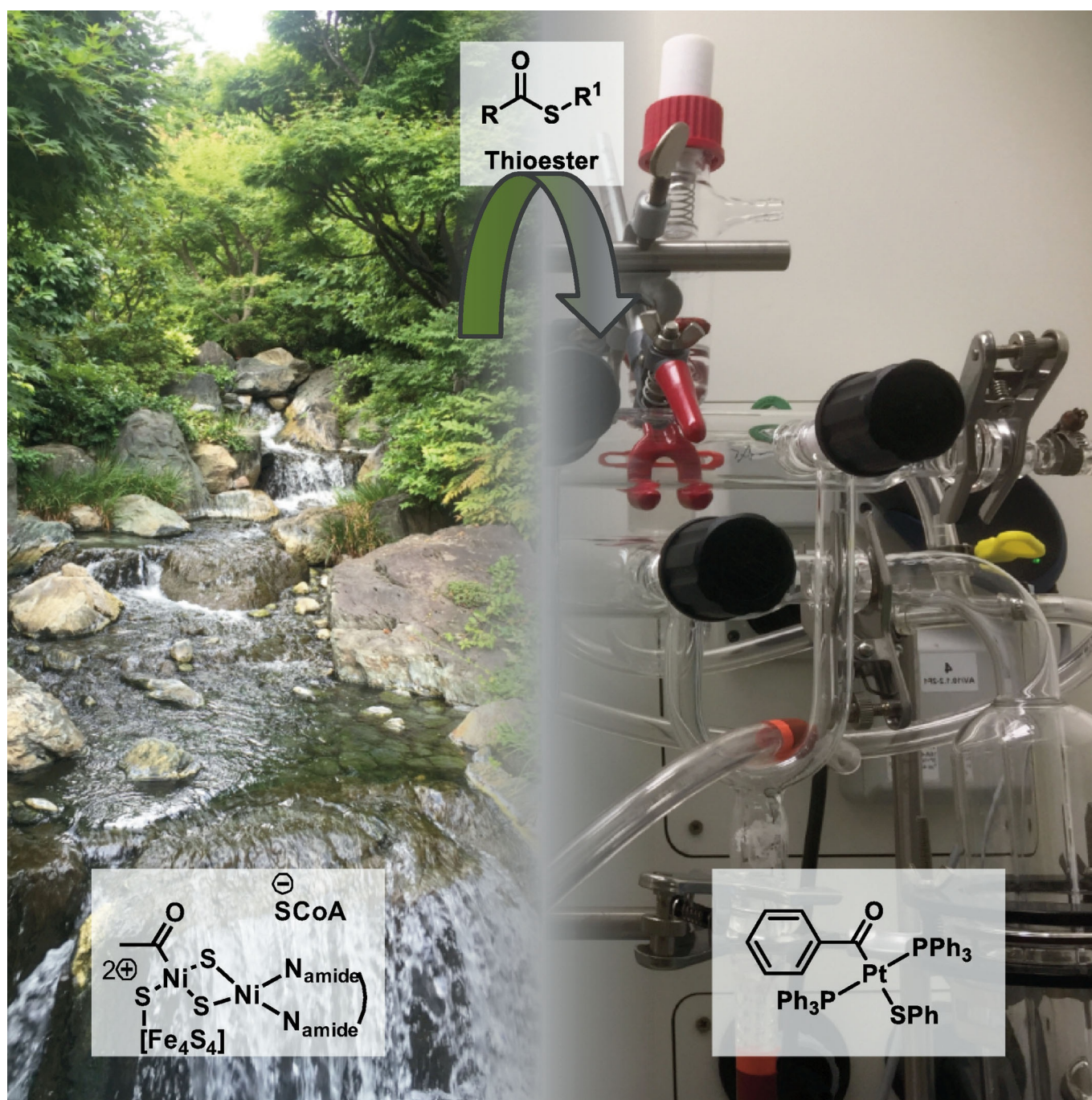


Synthetic Methods

Metal-Catalyzed Synthesis and Use of Thioesters: Recent Developments

Vera Hirschbeck^{+, [b]} Paul H. Gehrtz^{+, [a]} and Ivana Fleischer^{*, [a]}

Dedicated to Professor Štefan Toma on the occasion of his 80th birthday.



Abstract: While thioesters are common intermediates in biochemical processes, they are much less appreciated in organic synthesis, also compared to other carboxylic acid derivatives. However, their chemistry and reactivity is intriguing and diversified, reaching much further than the acyl substitution and aldol chemistry. Herein, we focus on metal-catalyzed reactions for the synthesis of thioesters as well as their transformations. Reactions such as thiocarbonylation, cross-

coupling, decarbonylation, allylic substitution or dual photo-redox/metal catalysis are discussed. On one hand, new atom economic methods allow for convenient synthesis of thioesters from well available starting materials. On the other hand, various synthetically important compounds can be synthesized due to the multifaceted reactivity of thioesters that we aimed to depict.

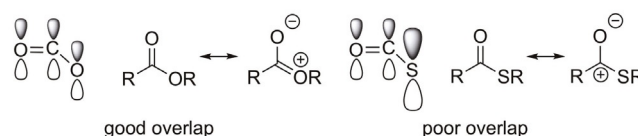
1. Introduction

According to certain hypotheses, thioesters played an important role in the development of life on earth—both as building blocks and source of energy.^[1] They are ubiquitous in biochemical processes, for example acetyl coenzyme A delivers the acetyl group into the citric acid cycle^[2] and other thioesters are involved in polyketide biosynthesis (Scheme 1).^[3] These processes benefit from the properties of the thioester moiety, which is more reactive towards nucleophiles than analogous oxoesters due to smaller orbital overlap between the sulfur atom and the carbonyl group. In addition, the better stabilization of the α -anion facilitates aldol chemistry.

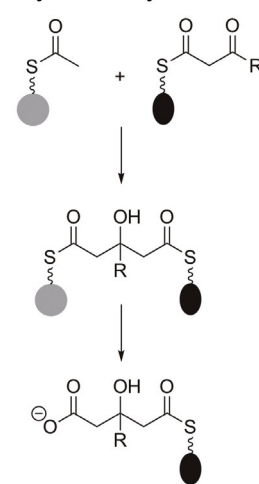
Consequently, thioesters became attractive also for synthetic chemists and numerous methods for their preparation and use were developed.^[4] Their reactivity is comparable to acyl chlorides. However, thioesters are bench stable and can be stored for a long time. Probably the most notable example of the use of thioesters is the native chemical ligation (NCL) method to connect peptides.^[5] In this process, a C-terminal thioester of one peptide molecule reacts with N-terminal cysteine, furnishing a thioester, which undergoes acyl transfer to the nitrogen atom. Thus, both the high nucleophilicity of a thiolate and high electrophilicity of the thioester are used in this reaction sequence.

Aside from acyl substitution^[4c] and aldol chemistry, valuable methods employing metal catalysts were developed for the synthesis and use of thioesters. This minireview aims to highlight selected recent methods for the synthesis of thioesters as well as their applications beyond the classical acyl substitution and aldol chemistry (Scheme 2). We focus on metal-catalyzed reactions, with most catalyst systems being homogeneous in nature. In general, several complications might arise when

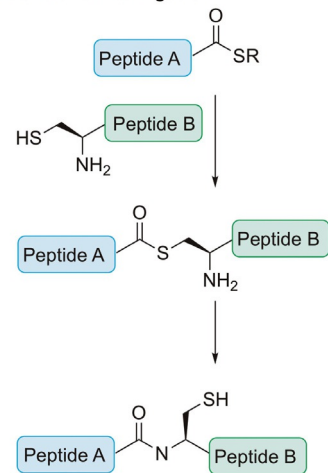
Oxo- vs. thioester



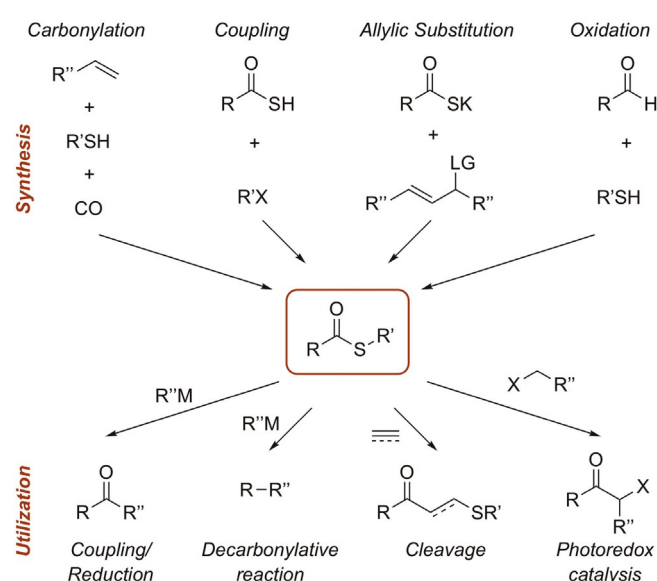
Polyketide biosynthesis



Native chemical ligation



Scheme 1. Thioesters: reactivity, use in biochemical processes and NCL.



Scheme 2. Overview of selected methods covered in this review.

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working with metals and sulfur-containing compounds. For example, sulfur compounds can act as poisons for transition metals due to the strong M–S bonds.^[6] In addition, side reactions can occur, often by radical pathways.^[7]

Among the methods available for the synthesis of thioesters, carbonylations of aryl halides and unsaturated compounds in the presence of thiols belong to the most atom- and waste economic procedures. They also constitute the most studied synthetic methods for thioesters using metal catalysts. More efforts were put into development of reactions employing thioesters.^[8] We will discuss cross couplings, reductions, decarbonylative and cleavage reactions and most recently developed applications in photoredox catalysis. Our aim is to depict the diversity of reactions involving thioesters, however, detailed mechanistic descriptions are beyond the scope of this review.

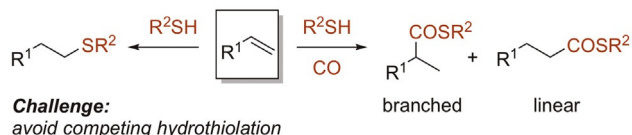
2. Syntheses of Thioesters

2.1. By carbonylations

Transition-metal-catalyzed carbonylation reactions are one of the most important synthetic strategies for organic synthesis as well as in industrial processes, which enables the generation of highly functionalized compounds from easily available starting materials.^[9] Carbonylation chemistry can also be used to generate thioesters, mainly from olefins or aryl halides.

2.1.1 Carbonylation of alkenes/alkynes

The direct carbonylation of olefins with thiols, is based on the comprehensive research of Walter Reppe (*1892, †1969).^[10] The so called Reppe-type thiocarbonylation, also referred to as hydrothioesterification, is a highly atom- and waste-economic method to synthesize complex thioesters in one step (Scheme 3). Although, this is an attractive alternative to more

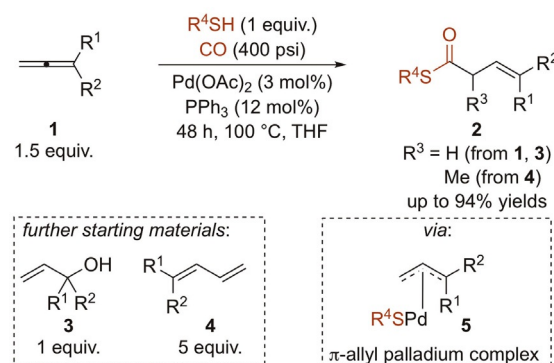


Scheme 3. Reppe-type thiocarbonylation of olefins and competing hydrothiolation.

common synthetic strategies, such as acylation of thiols, it is a poorly investigated research-field, since a number of obstacles need to be overcome. Working with free thiols is challenging, since they tend to oxidize and quickly generate the corresponding disulfides. Moreover, thiols are well known as catalyst poisons for late transition metals.^[6] Furthermore, the competing radical-mediated hydrothiolation of the olefin has to be avoided.^[11]

Most investigations on thiocarbonylation were realized by Alper and co-workers. They reported on thiocarbonylations with thiophenols and carbon monoxide in order to generate β,γ -unsaturated thioesters **2** either from allenes **1**,^[12] allylic al-

cohols **3**^[13] or 1,3-conjugated dienes **4**^[14] (Scheme 4). The best yields were obtained by using $\text{Pd}(\text{OAc})_2$ in combination with



Scheme 4. Thiocarbonylation of allenes (**1**), allylic alcohols (**3**) and 1,3-conjugated dienes (**4**) reported by Alper et al.^[12–14]

Vera Hirschbeck was born in Munich, Germany in 1991. She received her M.Sc. degree in organic chemistry at the University of Regensburg in 2015. In the same year she started her Ph.D. under the supervision of Prof. Dr. Ivana Fleischer and received the PhD Fellowship from Fonds der chemischen Industrie. Her research interest concerns different carbonylation reactions and the use of C1 building blocks as an alternative of toxic and gaseous carbon monoxide.



Paul Gehrtz was born in 1989 and grew up in Hamburg, Germany. He studied Biochemistry at the University of Bayreuth (B.Sc.) and Medicinal Chemistry at the University of Regensburg, and received his M.Sc. degree in 2015 with Prof. Dr. Ivana Fleischer in the field of homogeneous catalysis. He currently continues working in the same group as a PhD student. His research interests include the development of new transformations of thioesters as well as their synthetic methods.

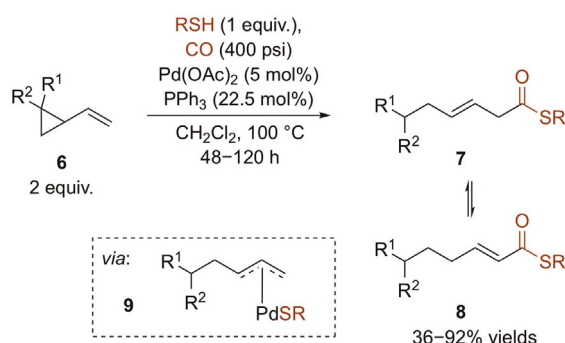


Ivana Fleischer studied chemistry at the Comenius University, Bratislava, Slovakia. She received her PhD from the University of Basel, Switzerland and then she joined the Leibniz-Institute for Catalysis in Rostock, Germany as postdoctoral fellow of the Swiss National Science Foundation. In 2013, she moved to University of Regensburg as a Liebig Fellow and group leader and in 2017 she was appointed as assistant professor at the University of Tuebingen. She is interested in development of new resource efficient catalytic transformations of olefins and carboxylic acid derivatives and in defunctionalisation reactions.



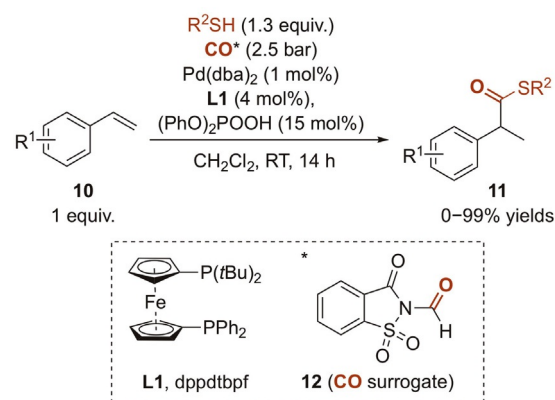
the monodentate PPh_3 ligand. Pd^{II} can be reduced to the catalytically active Pd^0 in the presence of the phosphine ligand and CO. The key intermediate for all three reactions is the π -allyl palladium complex **5**. In case of allylic alcohols **3** an acidic additive is required for protonation of the hydroxyl group, leading to elimination of water and generation of **5**. The reactions furnished excellent yields up to 94%. The high regioselectivity can be explained by the formation of the sterically less hindered π -allyl palladium complex **5**. A direct utilization of the Pd^0 source $\text{Pd}_2(\text{dba})_3 \cdot \text{CHCl}_3$ was also successful for all three synthetic strategies, albeit it led to slightly reduced yields. In 2001 Alper et al. reported the first enantioselective palladium-catalyzed thiocarbonylation of prochiral 1,3-conjugated dienes.^[15] They achieved good to excellent enantioselectivities up to 89% ee by using 5 mol% of $\text{Pd}(\text{OAc})_2$ combined with 10 mol% of (*R,R*)-DIOP (= (*R,R*)-*O*-isopropylidene-2,3-dihydroxy-1,4-bis-(diphenylphosphino)butane). The enantioselectivity-determining step was suggested to be the CO insertion to the π -allyl palladium complex **5**.

Later on, in 2009 Alper et al. investigated a ring opening thiocarbonylation of vinylcyclopropanes **6** catalyzed by 5 mol% $\text{Pd}(\text{OAc})_2$, 22.5 mol% PPh_3 with 400 psi CO at 100 °C, forming β,γ - or α,β -unsaturated thioesters (**7**, **8**, Scheme 5).^[16] The key intermediate is the π -allyl palladium complex **9**, which is converted to β,γ -unsaturated thioester **7**, and further on to its α,β -isomer **8**.^[17]



Scheme 5. Palladium-catalyzed thiocarbonylation of vinylcyclopropanes **6**.^[16]

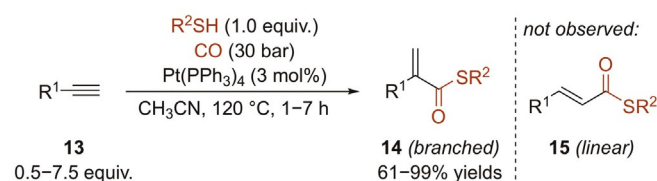
In 2016, Fleischer et al. reported the first chemo- and regioselective thiocarbonylation of vinyl arenes **10** by using 1 mol% of $\text{Pd}(\text{dba})_2$ and 2.5 bar CO in dichloromethane under ambient conditions (RT, 14 h) (Scheme 6).^[18] The chosen ferrocene ligand **L1** unifies steric demands, such as the required bite angle, and a unique electronic differentiation. The more electron-poor diphenylphosphino arm accelerates reductive elimination, whereas the relatively electron-rich di-*tert*-butylphosphino moiety favours the oxidative addition. The combination of both properties enabled a highly active and robust catalytic system, with an excellent group tolerance, providing isolated yields up to 99%. Moreover, the reaction proceeded in a highly regioselective fashion, exclusively generating the branched products **11**, even for sterically demanding groups in the *ortho*-position. Furthermore, the system was selective for



Scheme 6. Regioselective thiocarbonylation of vinyl arenes **10** using a CO surrogate.^[18] dba: dibenzylideneacetone; dppdtbpf: 1-diphenylphosphino-1'-(di-*tert*-butylphosphino)ferrocene.

terminal double bonds, whereas internal ones were not transformed because they cannot coordinate to the metal. Carbon monoxide was released from the CO-surrogate *N*-formylsaccharin (**12**, NFS) and the reaction was performed in a two-chamber pressure tubes developed by Skrydstrup.^[19] This concept was previously reported for the alkoxy carbonylation of olefins.^[20]

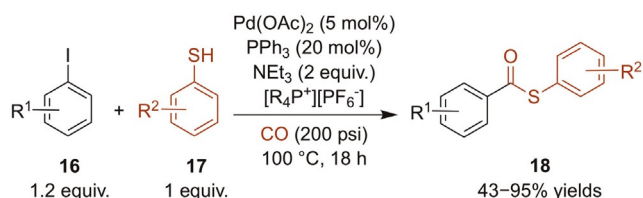
Furthermore, a few publications are available on the thiocarbonylation of alkynes. In 1997, Ogawa et al. described the platinum-catalyzed carbonylative addition of thiols to terminal alkynes **13**, generating α,β -unsaturated thioesters **14**.^[21] By using 3 mol% $\text{Pt}(\text{PPh}_3)_4$, 1-octyne was carbonylated with benzenethiol and CO (30 bar), providing regioselectively the branched product in 77% yield. Also aliphatic thiols were applied successfully (Scheme 7). El Ali et al. presented a thiocarbonylation of terminal alkynes in 2001, which was catalyzed by 1 mol% $\text{Pd}(\text{OAc})_2$ and 4 mol% of a bidentate ligand, dppb (= 1,4-bis(diphenylphosphino)butane) or dppp (= 1,3-bis(diphenylphosphino)propane).^[22] They were able to control the regioselectivity by a careful adjustment of the type of the ligand and the solvent, to obtain either the branched or the linear product.



Scheme 7. Carbonylation of alkynes providing α,β -unsaturated thioesters **14**.^[21]

2.1.2 Carbonylation of aryl halides

Starting from aryl halides, another carbonylation strategy is accessible in order to synthesize thioesters of benzoic acid derivatives **18**. In the pioneering work of Alper, the first Pd-catalyzed thiocarbonylation of aryl iodides **16** was published in 2008.^[23] A catalytic system consisting of 5 mol% $\text{Pd}(\text{OAc})_2$,

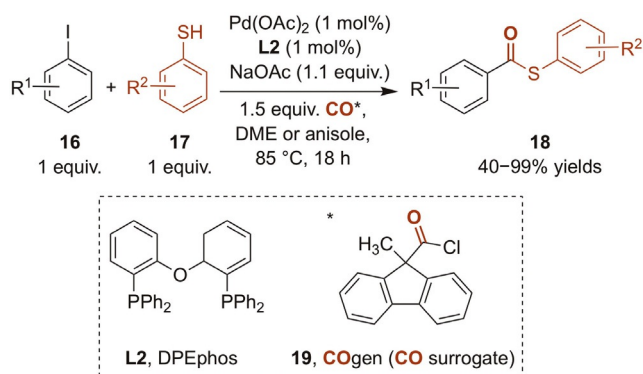


Scheme 8. First thiocarbonylation of aryl iodides reported by Alper et al.^[23]

20 mol % PPh₃ as ligand in combination with 2 equiv. of a base was used (Scheme 8). A phosphonium salt ionic liquid (PSIL) was chosen as a solvent, due to its excellent properties such as nonflammability, nonvolatility, chemical and thermal stability and most importantly the recyclability. The addition of hexane after the reaction enabled the formation of two phases with the palladium catalyst in the ionic liquid layer and the thioester in the organic phase. After recovery of the catalyst, another reaction cycle was possible without any loss of activity. The reaction featured a high functional group tolerance for both reaction partners, aryl iodides and thiols, whereas electron-withdrawing groups provided better yields than electron-donating substituents. Interestingly, also aliphatic thiols delivered good yields.

Similar to this work, Lei et al. reported a palladium catalyzed thiocarbonylation of aryl iodides with sodium thiolates and a thiol/THF solvent mixture, which was able to afford also sterically hindered thioesters, such as ArCOSTBu.^[24] Later on, in 2013 Skrydstrup et al. reported a Pd-catalyzed thiocarbonylation of aryl iodides **16** under milder reaction conditions, such as 1 mol % of Pd(OAc)₂, 1 mol % of ligand **L2** and especially by using only a stoichiometric amount of carbon monoxide, with NaOAc as a base and DME or anisole as a solvent.^[25] CO was generated *ex situ* from 9-methylfluorene-9-carbonyl chloride (**19**, COgen) in a two-chamber pressure tube (Scheme 9). A thorough solvent and ligand screening led to prevention of the competing thioether formation and enabled a highly efficient method to transform electron-rich and electron-deficient aryl iodides.

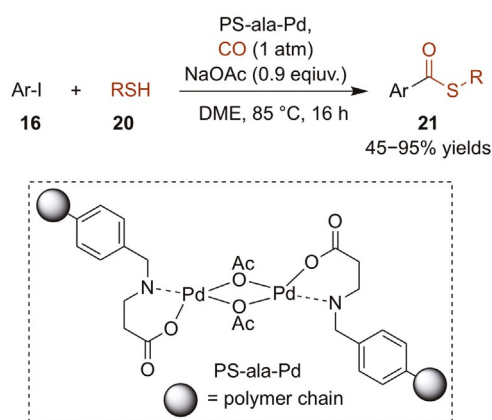
Based on this work, the same group presented a modified catalyst system for carbonylation of aryl, vinyl, and benzyl bro-



Scheme 9. Thiocarbonylation of aryl iodides using a CO surrogate reported by Skrydstrup et al.^[25] DPEphos: bis[(2-diphenylphosphino)phenyl] ether.

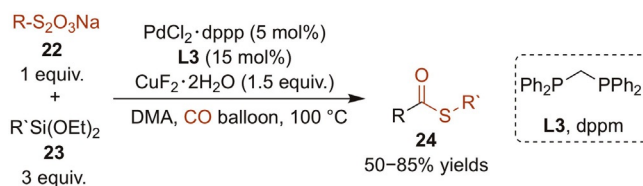
mides.^[26] The combination of 5 mol % Pd(PhCN)₂Cl₂ and 5 mol % of ligand Xantphos (=4,5-bis(diphenylphosphino)-9,9-dimethylxanthene) with NaOAc as a base in anisole at 120 °C for 18 h, proved to be the most efficient catalytic system, furnishing the desired products in up to 98% yield. An extensive substrate screening proved the applicability of the catalytic system.

A phosphine-free version for thiocarbonylation of aryl halides was reported by Islam et al. in 2014 (Scheme 10).^[27] They employed a polystyrene supported Pd^{II} complex (PS-ala-Pd), which can easily be recovered by filtration after the reaction and can be reused up to six times without significant loss in activity. Most importantly, PS-ala-Pd is an air and moisture stable catalyst, which can be synthesized from inexpensive and commercial available starting materials (β-alanine, chloromethylated polystyrene, Pd(OAc)₂) in a two-step procedure.



Scheme 10. Ligand free thiocarbonylation of aryl halides **16** using a polymer supported catalyst.^[27]

In 2016, Arndtsen et al. reported a method to introduce functionalized thiols, which are potentially incongruous for usual carbonylation conditions.^[28] The solution is a two-step strategy generating an aminopyridinium salt from aryl halide, CO and DMAP (4-dimethylaminopyridine) in the presence of 5 mol % Pd(PtBu₃)₂ via carbonylation, followed by a nucleophilic substitution with the corresponding thiol. In 2016, Jiang et al. developed a carbonylative Hiyama-type cross-coupling between thiosulfate **22** and the organosilicon reagent **23** catalyzed by PdCl₂-dppp (5 mol %) with additional ligand **L3** and only 1 atm of CO (Scheme 11).^[29] CuF₂ served both as oxidant



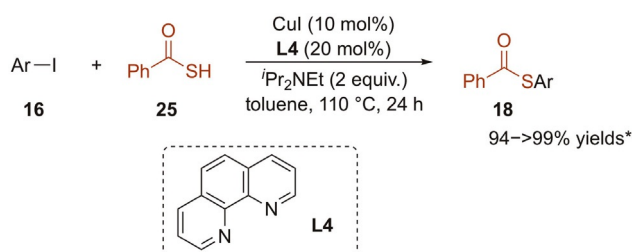
Scheme 11. Carbonylative cross-coupling of thiosulfates **22** and organosilicon compounds **20**.^[29] dppm: 1,1-bis(diphenylphosphino)methane. DMA: dimethylacetamide.

and fluoride source. The applied thiosulfate enabled a base-free, odourless and oxidizing-reagent-compatible reaction pathway. No reaction times were given.

In addition to Pd-based chemistry, the first nickel-catalyzed synthesis of thioesters from aryl iodides was presented by Iranpoor, Firouzabadi and co-workers in 2015.^[30] NiCl₂ was able to thioacetylate aryl iodides in the presence of a base, by using Cr(CO)₆ as a carbon monoxide source under air. Different functional groups and thiols were tolerated by the system providing good to excellent yields.

2.2. By coupling reactions

Thioesters can also be accessed by metal-catalyzed coupling of aryl electrophiles and thiocarboxylates. In early reports, reactive diazonium salts were used as electrophilic coupling partners without the need to employ any catalyst.^[31] Osuka and Suzuki established coupling of aryl iodides with copper thioacetate in HMPA (= hexamethylphosphoramide).^[32] This reaction was improved later by Sawada et al., who employed thioacetic acid (**25**), base and only 10 mol% of CuI in the presence of 1,10-phenanthroline (**L4**, Scheme 12).^[33] Several functional groups on aryl iodide were tolerated, but bromides were too unreactive under the reaction conditions. The method was later extended to thioacetates by Peññory, who also was able to employ it in a one-pot synthesis of heterocycles.^[34] A successful utilization of aryl bromides and triflates in this type of coupling was achieved by Lai et al. by using a catalyst system based on a Pd⁰ precursor and the bidentate phosphine Xantphos.^[35]

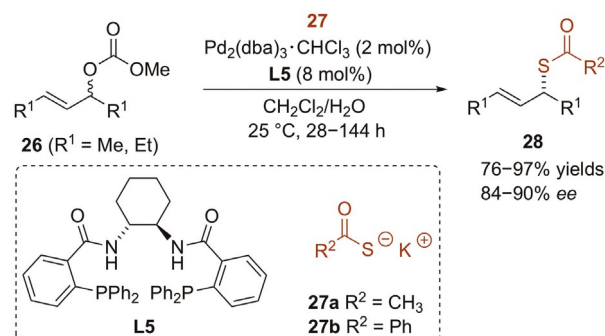


Scheme 12. Copper-catalyzed coupling of aryl iodides with thioacetate.^[33]
* The yields were determined by HPLC.

2.3. By allylic substitution

Since its discovery, the metal-catalyzed allylic substitution has become a valuable method to access a variety of chiral allylic substrates.^[36] Mostly, carbon nucleophiles have been studied, while heteroatom-based nucleophiles are more challenging.^[37] The difficulties of these transformations are associated with the properties of the O, N, S-nucleophiles, such as better leaving group ability leading to reversible reactions or their propensity to bind to metals as ligands. However, several successful transformations using heteroatoms were developed over the years. Sinou and co-workers found that it is possible to use thioacetate as nucleophile in the allylic substitution using a Pd⁰ precursor and a phosphine ligand.^[38] An asymmetric ver-

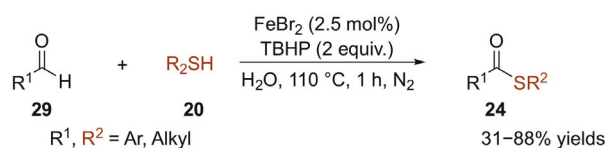
sion was developed by Gais et al. (Scheme 13).^[39] Employing the Trost ligand **L5**, they achieved good yields and ee values in the transformation of racemic cyclic and acyclic allylic carbonates **26** with thioacetate **27a** and thioacetate **27b**. Several carbonates and also acetates were subjected to kinetic resolution. A competing formation of allylic alcohols was observed when cyclic carbonates reacted with thioacetate due to the presence of water in the solvent mixture.



Scheme 13. Asymmetric allylic substitution using thiocarboxylates shown for acyclic substrates.^[39]

2.4. By oxidative transformation of aldehydes and methyl arenes

Oxidative coupling of aldehydes with thiols in the presence of an appropriate oxidant/catalyst and the related conversion of methyl arenes provide the corresponding thioesters. The transformation of aldehydes has been achieved with various catalysts, such as N-heterocyclic carbenes^[40] or by using radical initiators.^[41] Among the metal-based catalysts, Lee and co-workers employed CuCl₂^[42] and later FeBr₂^[43] in the presence of TBHP (= *tert*-butyl hydroperoxide) as oxidant (Scheme 14). A variety of aromatic and aliphatic aldehydes **29** and thiols **20** furnished thioesters **24** in good to high yields, when the reaction was performed in water at 110 °C. The authors suggested a mechanism for the iron-catalyzed reaction including Fe^{III}- and acyl radical species.



Scheme 14. Iron-catalyzed oxidative coupling of aldehydes and thioesters.^[43]

Similarly, it was shown that iron and copper catalysts can be used to convert toluene derivatives and thiols to the corresponding thioesters.^[44] It was suggested and confirmed by competition experiments that the methyl group is first oxidized to an aldehyde. Interestingly, no metal catalyst was needed in the work of Lee et al., who just used disulfides and an oxidant.^[45] They proposed the initial formation of a thio-

ether and its subsequent oxidation. Related approaches to thioesters were developed starting from α -oxocarboxylic acids via decarboxylative pathways.^[46]

3. Uses of Thioesters

3.1. In coupling reactions

The reaction between selected organometallic reagents (d^1 -type) and some carboxylic acid derivatives (a^1 reagents) can selectively furnish ketones (Scheme 15). The general text book known challenge is that the formed ketones are usually more reactive and alcohols are often formed as main products. Compounds such as Weinreb amide^[47] **29** or Staab–Jost imidazole^[48] **30** are examples of compounds that furnish ketones with Grignard reagents in the absence of any catalyst due to the formation of a chelate intermediate. Thioesters **21** constitute attractive alternatives to these compounds. While the significantly more activated *S*-(2-pyridyl)thioesters, which presum-

ably can also form a stoichiometric chelate,^[49] fall partially into the area of classical a^1 reagents, other thioesters can be employed in the presence of a transition metal catalyst. In this case, two mechanistic scenarios are plausible: low-valent transition metal species undergo oxidative addition into the C–S bond (Scheme 15, A) or the thioester is the acceptor in a single electron transfer (SET) event (Scheme 15, B).

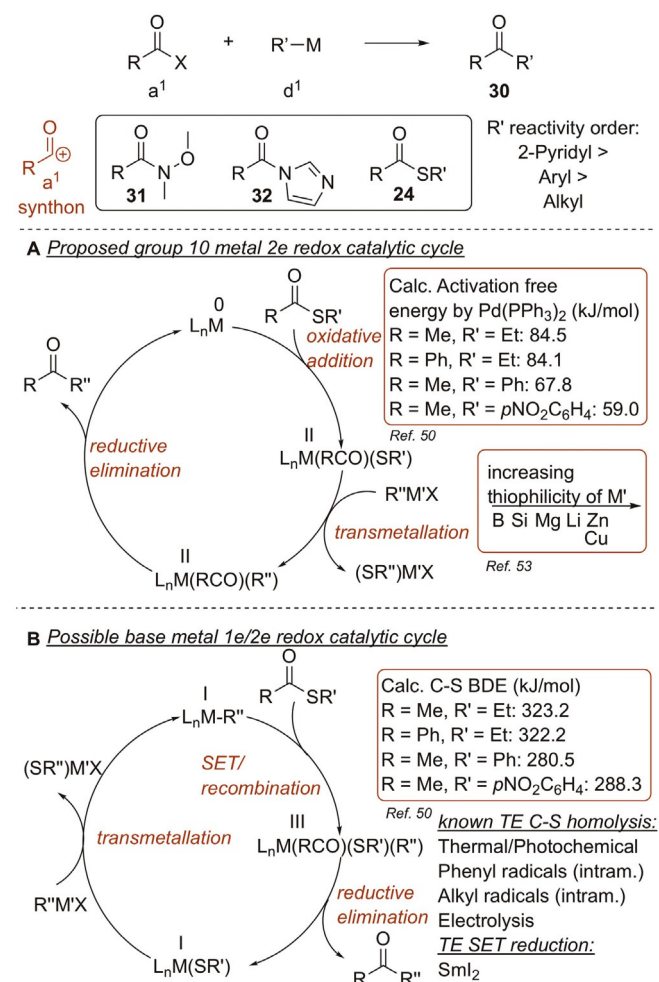
It should be noted that a DFT-based computational comparison of C–S bond dissociation energies (BDE's) versus C–S bond activation energies by a Pd^0 model complex of various thioesters has shown only a weak correlation.^[50] It was then proposed, at least for the more noble transition metals, that a general cross-coupling catalytic mechanism is followed. Transmetalation liberates a thiolate from the complex, which is followed by reductive elimination to regenerate the low-valent transition metal complex and release the product.

In the absence of a transition metal catalyst, only organocuprates were shown by Rosenblum to react cleanly with thioesters to give ketones,^[51] presumably via a Cu^{III} intermediate,^[52] and not an addition/elimination sequence. This shows that effects other than the extent of C–M bond polarization of the organometallic coupling partner are relevant for a successful reaction of thioesters. On the other hand, it is also clear that highly polar organometallic species (RLi , $RMgX$) should react with the formed ketones, which makes them disfavoured in ketone synthesis from thioesters in any case. Other effects could for example be the thiophilicity^[53] of the stoichiometrically employed metal to influence the thermodynamics of the transmetalation step. Liberation of the thiolate during the course of a reaction is the central problem when thioesters are employed as acylating agents since effective sequestration of this species must take place during the reaction to avoid a poisoning effect, a problem highly pronounced with late, thiophilic transition metals. In the following sections, we will discuss selected examples on successful utilization of thioesters in ketone synthesis in metal-catalyzed reactions.

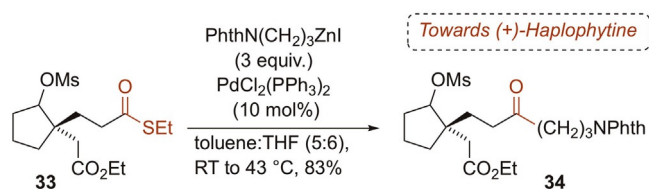
3.1.1 Catalysis with Pd

In 1998, Fukuyama and co-workers reported a seminal Pd-catalyzed ketone synthesis from *S*-ethyl thioesters employing primary alkyl- and aryl zinc reagents (1.5 to 3.0 equivalents)—now known as the Fukuyama coupling.^[54] As expected with organozinc reagents, the functional group tolerance is excellent both in nucleophile as well as electrophile. The reaction proceeded with 5–10 mol% $PdCl_2(PPh_3)_2$ in THF or toluene at RT in 5 min to 3 h, giving isolated yields ranging from 50 to 99%. This reaction has been applied by Fukuyama in his synthesis of (+)-Haplophytine (Scheme 16).^[55] The thioester group was carried through two synthetic steps untouched, and then engaged in coupling with a functionalized alkylzinc iodide reagent.

This reaction has been found in additional reports to proceed without ancillary ligand^[56] or even with Pd/C ,^[57] from which was inferred by further experimentation that both heterogeneous and homogenous catalytic cycles are operative in the Pd/C -catalyzed Fukuyama coupling.^[58] The reaction has since then been expanded to accommodate secondary orga-

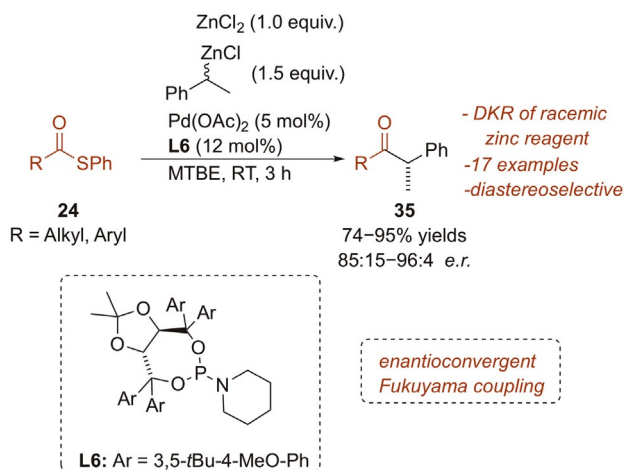


Scheme 15. Thioesters are alternatives to classic reagents for the a^1 synthon in the synthesis of ketones. Activation with group 10 metals is proposed to follow pathway A, whereas activation with low-valent base metals could follow pathway B. BDE: Bond dissociation energy. SET: Single electron transfer. TE: Thioester.



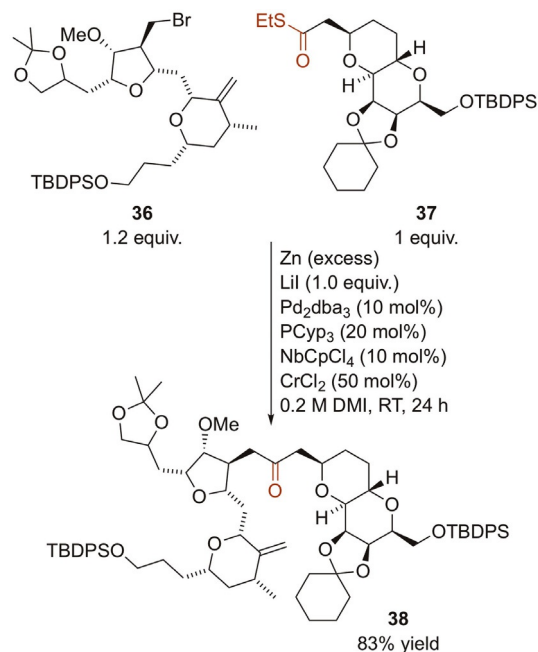
Scheme 16. Fukuyama's total synthesis of (+)-haplophytine employs the Fukuyama coupling reaction to give a synthetic fragment **34** in the early stages.^[55] Ms: Mesyl, PhthN: Phthalimide.

nozinc coupling partners.^[59] Notably, a highly stereoconvergent catalytic system, which forms tertiary stereocenters from racemic benzyl zinc reagents and thioesters **24**, has been disclosed by Maulide's group (Scheme 17).^[60] They used an enantiopure chiral phosphoramidite ligand (**L6**) as the stereoinducing element to affect a dynamic kinetic resolution (DKR) of the zinc reagent.



Scheme 17. Maulide's synthesis of tertiary stereocenters by an enantioconvergent Fukuyama reaction.^[60] MTBE: methyl-*tert*-butyl ether.

Kishi pointed out that in the late stages of complex molecule synthesis, acylation chemistry should occur with optimally 1:1 stoichiometry of the coupling partners due to their high value with an additional high functional group tolerance. Thus, a one-pot Fukuyama reaction was developed by the group employing Pd_2dba_3 (5 mol%) and $\text{P}(\text{Cyclopentyl})_3$ (10 mol%) in DMI (1,3-dimethyl-2-imidazolidinone) at RT for 1 day (Scheme 18).^[61] The aim was to extend the methodology into a macroketocyclization method in the synthesis of Eribulin. It should be noted that this reaction is not only one-pot but it also belongs to the class of cross-electrophile couplings. Presumably, the alkyl bromide is converted to the alkyl iodide in the presence of LiI, followed by insertion of Zn dust (activated by stoichiometric triethylsilyl chloride). Either the lithium iodide or lithium bromide present could additionally facilitate zinc insertion, as this is a well-known salt effect in organozinc preparations.^[62] Depending on the exact substrate class, a combination of various metal-containing additives (Co-Phthal-

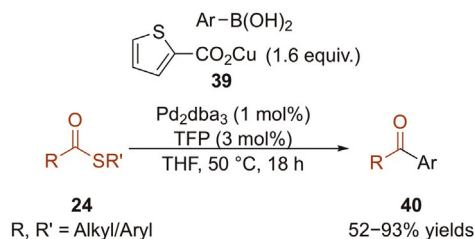


Scheme 18. Fukuyama-type cross-electrophile coupling of complex thioesters and alkyl bromides reported by Kishi.^[61] TBDPS: *Tert*-butyl-diphenylsilyl, dba: dibenzylidenacetone, Cyp: cyclopentyl, Cp: cyclopentadienyl.

cyanine or NbCpCl_4 as SET mediators, CrCl_2 as a de-aggregating species and alkyl radical trap) in substoichiometric amounts were necessary for consistent results with the most challenging substrates. The cross-electrophile coupling of fragments **36** and **37** shows an impressive functional group tolerance and insensitivity to detrimental aggregate effects, which can occur when coupling fragments with large molecular weights and different solubility behaviours.

These studies on intermolecular reactions were followed by a synthesis of the microtubule-modulatory chemotherapeutic agent Eribulin (HalavenTM, against breast cancer) featuring a macroketocyclization at a remarkably high concentration of 27 mM in DMI/THF. However, (super-)stoichiometric amounts of Pd, ligand and additives were required for attaining an acceptable yield.^[63] Other catalytic (Fe, Ni) variants of this cross-electrophile coupling have been published by Kishi using *S*-(2-pyridyl)thioesters^[64] and subsequently applied in the total synthesis to some members of the halichondrin natural product series.^[65]

Another milestone in this area is the Liebeskind–Srogl coupling, which was disclosed in 2000 (Scheme 19).^[66] Here, aryl boronic acids were employed to render the transformation especially facile with respect to the stoichiometric coupling partner. In this case however, Pd–S bond activation in the transmetalation (or maybe C–S activation in the oxidative addition) step must be facilitated by an additional stoichiometric Cu^I source (**39**). The copper-bound carboxylate anion is proposed to activate the boronic acid reagent; hence the transmetalation step is thought to encompass a ternary complex. Thirteen preparative examples with diverse functional groups were shown in isolated yields up to 93% using low catalyst loading in THF at 50 °C for 18 h.



Scheme 19. The Pd-catalyzed, Cu-mediated Liebkund-Srogl coupling of arylboronic acids with thioesters.^[66] TFP: tris(2-furyl)phosphine.

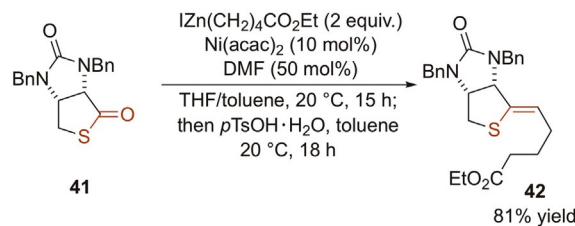
Since then, the scope of stoichiometric coupling partners has been extended towards 9-BBN-based alkylboronates,^[67] organostannanes^[68] and organoindium^[69] reagents. The indium variation does not require a stoichiometric Cu^I source, which highlights again the requirement of sufficient thiophilicity of the stoichiometric transmetalating agent. A significant advance in this area has been made by Villalobos and the original inventors of the reaction, which now allows a Cu-catalyzed (5 mol%) aerobic variation of the reaction by fine tuning the thioester moiety to an S-acyl thiosalicylamide.^[70] An exhaustive description of the advances and variations in this area is not the aim of this chapter; the reader is referred to a specialized review.^[71]

Similarly non-polarized stoichiometric coupling partners are organosilanes, which were used (1.1 to 1.2 equivalents) by Van der Eyckens and co-workers in a thioester-Hiyama coupling.^[72] This method requires stoichiometric activation by fluoride (2.0 equiv) as well as copper iodide (1.0 equiv) in THF at 60 °C combined with Pd(PPh₃)₄ (5 mol%) to give products ranging from 79 to 98% isolated yields in 0.5–3.5 h. The other end of the polarity scale concerning the organometallic partner was covered by the same group by developing a thioester-Kumada coupling,^[73] where slow addition of the Grignard solution (1.2 equivalents) by a syringe pump at room temperature was key to success combined with a catalytic system consisting of Pd(dba)₂ (5 mol%) and tris(2-furyl)phosphine (TFP) (10 mol%) in a THF/*N*-methylpyrrolidone (NMP) solvent mixture with isolated yields ranging from 23–96%.

3.1.2 Catalysis with Ni

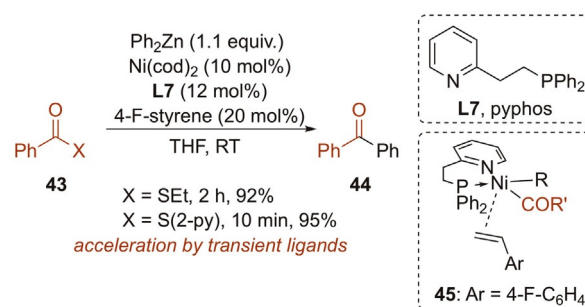
Seki and Shimizu were the first to report a mild Ni-catalyzed variant of the Fukuyama reaction by coupling (functionalized) aryl- and alkylzinc iodides with non-activated S-alkyl thioesters in 2002.^[74] Interestingly, the reaction proceeds with simple Ni(acac)₂ (10 mol% loading) without the need for ancillary ligands. Various alkyl/alkyl, aryl/alkyl and aryl/aryl ketones were obtained using this method in yields ranging from 33 to 89%. This method was applied by the authors to obtain a late stage vinyl sulfide intermediate **42** in the synthesis of (+)-biotin by a one-pot Fukuyama coupling/condensation sequence (Scheme 20).

Zhang and Rovis reported, rather as a side note in a study on the reactivity of acyl fluorides, a Ni-catalyzed variant of the Fukuyama coupling employing thioesters with diphenylzinc at



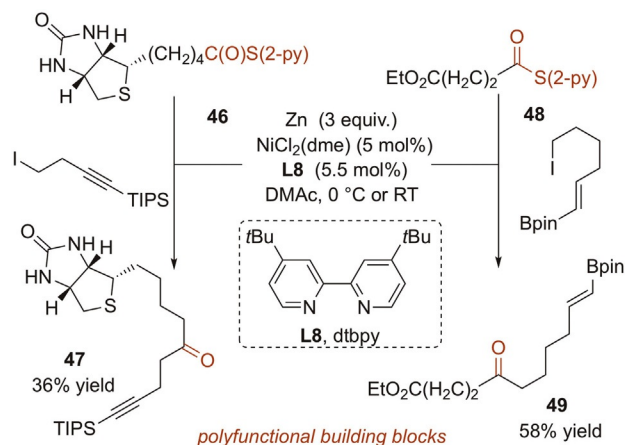
Scheme 20. Synthesis of the alkenyl sulfide intermediate **42** by a Ni-catalyzed Fukuyama coupling of thiolactone **41**.^[74] acac: acetylacetonate, pTsOH: *para*-toluenesulfonic acid.

room temperature with short reaction times.^[75] We think the comparison in reaction times between carboxylic acid derivatives conducted by the authors is highly instructive and highlights the differences in thioester activation. The reaction times with S-(2-pyridyl)thioesters are significantly shorter, hinting at their different intrinsic reactivity and/or ability of the corresponding thiolates to transmetalate at a transition metal center. Key to a successful reaction was the use of the highly active bidentate P,N-ligand **L7** together with 4-fluorostyrene as a transient π -acidic ligand to facilitate reductive elimination from complex **45** (Scheme 21).



Scheme 21. Ni-catalyzed coupling of thioesters using electron-withdrawing transient ligands.^[75] cod: 1,5-cyclooctadiene, pyphos: 2-[2-(diphenylphosphanyl)ethyl]pyridine.

The first Ni-catalyzed cross-electrophile coupling of a S-(2-pyridyl)thioester with an alkyl iodide was described by Mukaiyama et al., who employed NiCl₂ (10 mol%) in DMF at 50 °C together with three equivalents of zinc dust to afford a dialkyl ketone in 46% yield after 5 h, whereas reactions with the corresponding carboxylates were usually higher yielding.^[76] Weix and Wotal significantly improved this reaction by employing NiCl₂(dme)/L8 as the catalytic system with Zn as the reductant (Scheme 22).^[77] The method was applied to generate valuable polyfunctional compounds, such as “clickable” Biotin derivative **47**. In contrast to other organometallic functionalizations of Biotin, N-protection was not necessary, and the modest yield should be seen in relation with the reduced step count. The ketone linkage is an alternative to the usual amide bonds found in biotin derivatives, which are susceptible to hydrolysis by biotinidase. The possibility to employ downstream cross-coupling chemistry can also be introduced with no detriment as shown by the synthesis of a keto-boronate ester **49**.

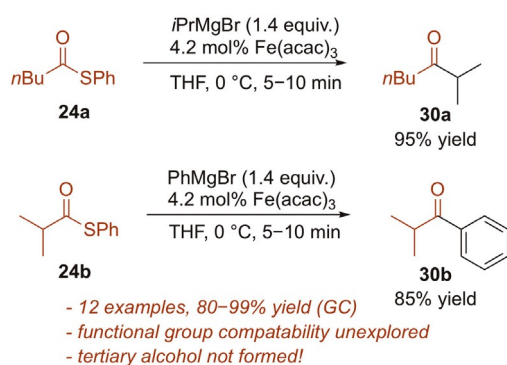


Scheme 22. Ni-catalyzed cross-electrophile coupling for the synthesis of highly functional ketones amenable to click-chemistry (**47**) or cross-coupling (**49**).^[77] dtbpy: 4,4'-di-*tert*-butyl-2,2'-bipyridine, dme: 1,2-dimethoxyethane, DMAc: dimethylacetamide.

3.1.3 Catalysis with Fe

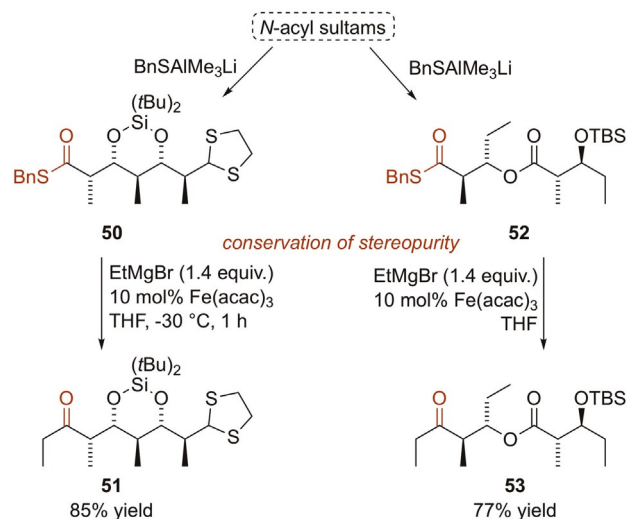
In 1985, Marchese and co-workers presented an $\text{Fe}(\text{acac})_3$ -catalyzed (4.2 mol%) synthesis of ketones from *S*-phenyl thioesters by reaction with Grignard reagents (1.4 equiv) at 0 °C in THF for 5–10 min (Scheme 23).^[78] The obtained yields in 8 mmol scale were very good to excellent for primary, secondary and aromatic Grignard reagents, with the same variation possible in the thioester structure. Notably, the formation of a tertiary alcohol as a byproduct was not reported in this finding, meaning that the coupling reaction must be faster by several orders of magnitude. The reaction mechanism is unknown, but it stands to reason that low-valent ate species could be formed from a Grignard reagent/ $\text{Fe}(\text{acac})_3$ mixture.^[79] Despite the operational advantages of this method, the use of Grignard reagents limits the functional group tolerance that has become important in the synthesis of complex structures or generally for late stage functionalization approaches.

In addition, Oppolzer, de Brabander and co-workers could show that *S*-benzyl thioesters can also be applied using the same methodology (albeit with 10 mol% loading of $\text{Fe}(\text{acac})_3$, and slightly different conditions).^[80] More importantly, enantiopure enolizable α -substituted thioesters were converted to ke-



Scheme 23. Convenient iron-catalyzed coupling of *S*-aryl thioesters with Grignard reagents shown by Marchese et al.^[78] acac: acetylacetonate.

tones with no erosion in enantiopurity observed. *S*-Benzylic thioesters were chosen as substrates due to easy access from *N*-acylsultam chiral auxiliaries by ate-complex mediated thiolysis. However, the use of methyl-, vinyl- and allyl magnesium halides primarily led to the corresponding tertiary alcohols. This strategy was elaborated to produce more complex building blocks, for example, an enantiopure intermediate **53** in the synthesis of (–)-serricorole, which contained an ester as well as a silyl ether-protected alcohol (Scheme 24).

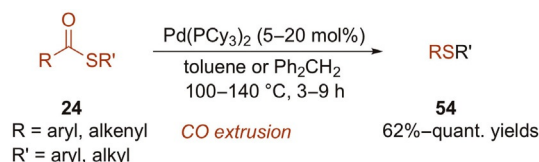


Scheme 24. Synthesis of functionalized, stereopure fragments **51** and **53** employing the Marchese-Fiandese cross-coupling. The thioesters were obtained by thiolytic cleavage of the amide-based chiral auxiliary.^[80]

3.2. In decarbonylative reactions

The extrusion of gases in concert with C–C or C–X bond formation as a preparative approach has recently gained attention since the parent carboxylic acid(s) (derivatives) are widely abundant chemicals. As opposed to CO_2 extrusion/cross coupling sequences,^[81] CO extrusion is more problematic since a potential transition metal catalyst poison is generated.^[82] Hence, the exact method of reaction setup can become an important factor for success of such reactions. Additionally, CO extrusion from thioesters has received little attention compared to the corresponding process from esters.

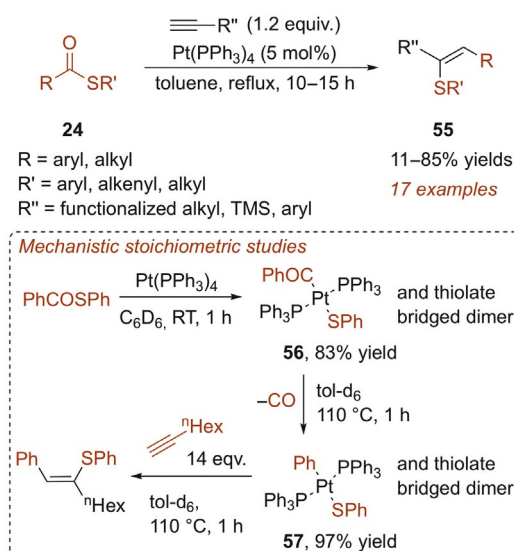
In the late 1980s, Yamamoto and co-workers reported a decarbonylation of thioesters **24** with catalytic $\text{Pd}(\text{PCy}_3)_2$ giving the thioether products **54** in high yields (98–100%, by GC; Scheme 25).^[83] Whereas *S*-phenyl thioesters were converted with 5 mol% of the catalyst in 3 h at 100 °C in toluene, an *S*-methyl thioester required a temperature of 140 °C in diphenylmethane, a loading of 20 mol% and 9 h reaction time to achieve a quantitative yield of methylphenyl sulfide. Reactions of unsaturated thioesters proved more challenging with Pd-catalysis, and several minor products were also detected. In this case, stoichiometric $\text{RhCl}(\text{PPh}_3)_3$ proved effective at 100 °C in DMF for 3 h for quantitative generation of vinyl sulfides. Kambe and Kuniyasu were able to show that appropriately placed lone pair electrons (from chloride, methoxy, or phenyl-



Scheme 25. Pd-catalyzed decarbonylation of thioesters.^[83]

sulfide substituents) in α,β -unsaturated thioesters greatly facilitated the oxidative addition, and hence, the overall catalytic cycle of decarbonylation to yield vinyl sulfides by $\text{Pd}(\text{PPh}_3)_4$ -catalysis (5 mol%) in toluene at reflux for 1 h.^[84]

Kurosawa, Kambe and Kuniyasu employed a $\text{Pt}(\text{PPh}_3)_4$ -catalyzed (5 mol%) decarbonylative fragmentation of thioesters to generate *trans*- $\text{Pt}(\text{SAr})(\text{Ar}')(\text{PPh}_3)_2$ complexes, which reacted with alkynes in a carbodithiolation/reductive elimination sequence to provide vinyl sulfides **55** in variable yields from 11–85% in refluxing toluene for 10–15 h (Scheme 26).^[85] Impor-

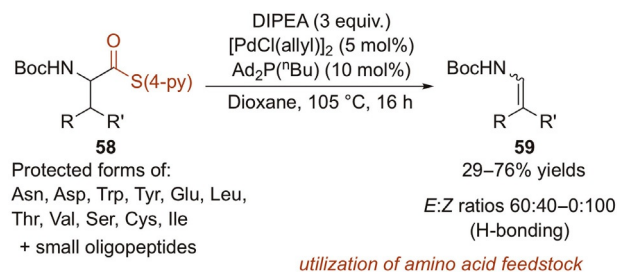


Scheme 26. Decarbonylative carbodithiolation of terminal alkynes.^[85]

tantly, the authors were able to show that while Pd was competent for decarbonylation of thioesters, it did not catalyze the hydrothiolation reaction. Moreover, isolated organometallic intermediates from stoichiometric experiments underwent the expected transformations, such as the thermal decarbonylative decomposition (1 h, 110 °C, toluene) of complex **56** to the key intermediate **57**. This concept has been extended towards (benzo)furan- and pyridyl-containing thioesters, and internal alkynes by the same group.^[86]

Another type of decarbonylative transformation was developed by Lee and co-workers, who used potassium thioacetate as the sulfide source to obtain unsymmetrical diaryl sulfides from aryl bromides and iodides with $\text{Pd}(\text{dba})_2$ (10 mol%), dppf (14 mol%) and K_3PO_4 in toluene/acetone at 70 °C for 3 h, subsequently for 6 h at 110 °C.^[87]

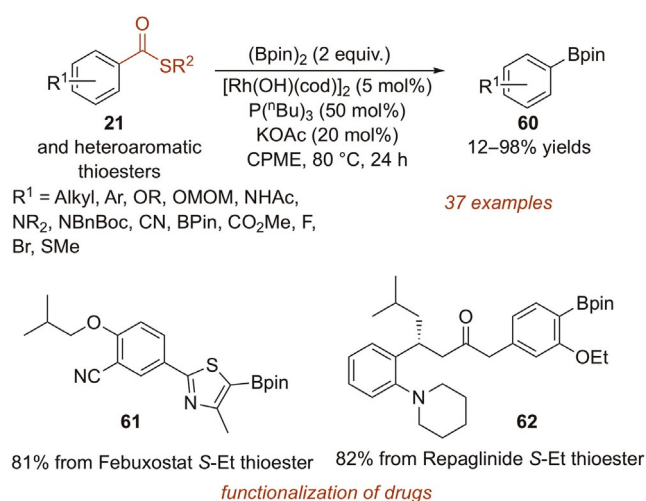
A desulfenylative-decarbonylative approach was employed by Skrydstrup to produce enamides **59** from various protected α -amino acid thioesters **58** using $[\text{PdCl}(\text{allyl})]_2$ (5 mol%), $n\text{BuPAD}_2$ (10 mol%) and DIPEA (3 equiv) in dioxane at 105 °C for 16 h (Scheme 27).^[88] The products were obtained in isolated



Scheme 27. Decarbonylative utilization of amino-acid derived thioesters **58** to provide enamides **59** developed by Skrydstrup.^[88]

yields ranging from 29 to 76% with variable *E:Z* ratios (the *E* product being preferred when no hydrogen bond stabilization can occur). Noteworthy, strong ligand effects were observed, with $n\text{BuPAD}_2$ being superior to nine other phosphine ligands. A short study of the leaving group effect was conducted, and while *S*-(4-pyridyl) substituted thioesters gave a similar yield compared to the *S*-(2-pyridyl) analogues, the stereoselectivity in the former case was better. The *S*-phenyl thioester gave a slightly decreased yield compared to the *S*-heteroaromatic thioester.

A desulfenylative-decarbonylative-borylative transformation was shown by Hosowa and Niwa in 2017 (Scheme 28).^[89] *S*-Ethyl aryl thioesters **21** underwent coupling with (Bpin)₂ using catalyst system based on $[\text{Rh}(\text{OH})(\text{cod})]_2$ (5 mol%), $\text{P}(n\text{Bu})_3$ (50 mol%), KOAc (20 mol%) in etheric solvent at 80 °C for 1 d. Particular CO poisoning effects were not noted, however, the success of the reaction depended strongly on the presence of KOAc. Highlights of this appealing transformation are success-

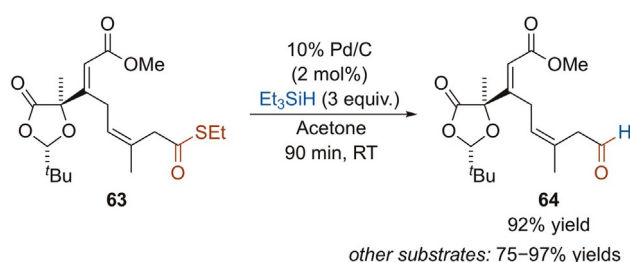


Scheme 28. Decarbonylative-desulfinitative borylation of aromatic thioesters.^[89] pin: pinacolato, cod: cyclooctadiene, CPME: cyclopentyl methyl ether.

ful borylations of febuxostat and repaglinide *S*-ethyl thioesters in 81 and 82% yield, respectively. From an earlier report on desulfenylative borylation by the authors using a similar system, it was argued that electron-rich trialkylphosphine ligands might facilitate the remarkable Rh-SR to Rh-Bpin transmetalation.^[90]

3.3. In reductions

The selective reduction of carboxylic acid derivatives to aldehydes constitutes a challenging task, since usually the aldehyde is more reactive in the reduction step. Among metal-catalyzed reactions, there are only few examples for the transformation of thioesters.^[91] In 1990 Fukuyama and co-workers developed a Pd-catalyzed reduction of thioesters to aldehydes by using triethylsilane (Scheme 29).^[92] They used 10% Pd/C as cat-

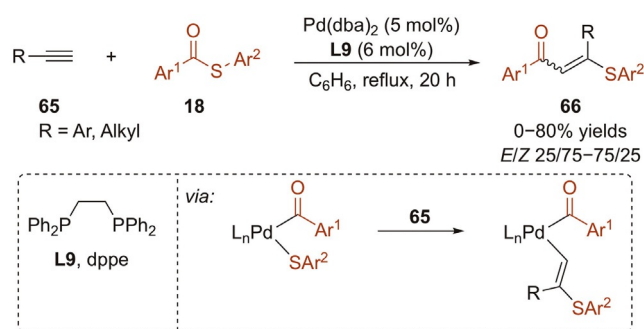


Scheme 29. Chemoselective Fukuyama reduction of thioesters to aldehydes.^[92]

alyst for the conversion of several thioesters to the corresponding aldehydes in good to high yields. Noteworthy, also highly functionalized substrates, such as **63** containing ester moieties and double bonds selectively furnished the desired product **64** in 92% yield. The authors later extended the methodology to differently protected α -amino acid derivatives, which were converted without racemization and with unchanged protecting groups such as Boc, Cbz and Fmoc.^[93] The reaction is sometimes referred to as Fukuyama reduction. Recently, Polyzos and Lupton developed its variation under flow conditions using a polymer supported Pd-catalyst.^[94] They also integrated the transformation into a multistep procedure starting from acyl chlorides and included purification strategies, thus increasing its practicality.

3.4. In C(O)–S bond cleavage and transfer reactions

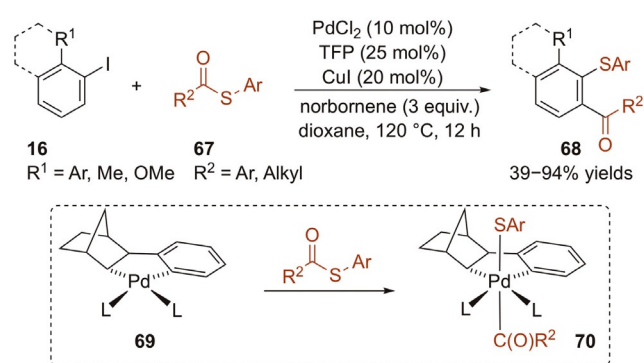
In this section, mechanistically distinct reactions involving the cleavage of C–S bond are discussed. The transformations are characterized by the transfer of both acyl and thiol moieties to the reaction partner, which might be an unsaturated compound, functionalized arene or a heterocycle. Kambe, Kuniyasu and co-workers realised a Pd-catalyzed thioacylation of terminal alkynes **65** with thiobenzoates **18** (Scheme 30).^[95] The catalyst system consisted of a Pd(dba)₂ precursor and the bidentate ligand dppe (**L9**, = 1,2-bis(diphenylphosphino)ethane), which promoted the formation of one product regioisomer **66**



Scheme 30. Palladium-catalyzed thioacylation of alkynes.^[95]

as a *E/Z* mixture. The authors obtained low yields with thiol ester of trifluoroacetic acid and found Pt(PPh₃)₄ to be a superior catalyst. This is in contrast to their previous work using this catalyst and thiobenzoates, where they observed a decarbonylation.^[86d] A mechanism was suggested, which starts with an oxidative addition of thioester onto Pd⁰. Alkyne inserts then into the Pd–S bond and the subsequent reductive elimination yields the product. Moreover, a related addition of thioesters to norbornenes was reported^[96] and one example for the reaction with allenes was shown.^[97]

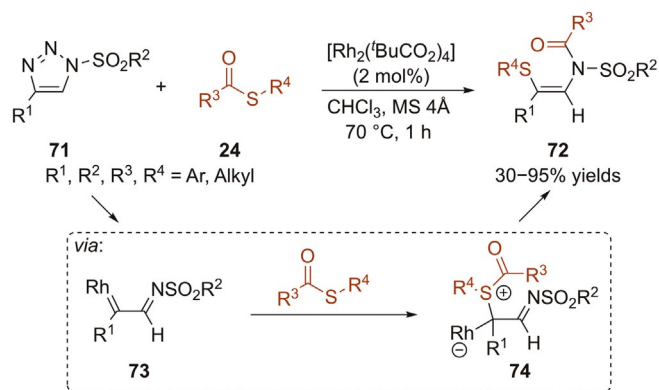
A Catellani-type^[98] synthesis of 2-(arylthio)aryl ketones **68** from aryl iodides and thioesters via the C(O)–S bond cleavage was developed by Gu et al. (Scheme 31).^[99] These transforma-



Scheme 31. Catellani-type reaction of thioesters catalyzed by a Pd/norbornene/Cu system.^[99]

tions are in general catalyzed by Pd precursors in the presence of norbornene and they allow for concurrent functionalization of aryl iodides in *ortho*- and *ipso*-positions. In the herein discussed case, a successful transformation of various *ortho*-substituted aryl iodides **16** and aryl thioesters **67** was achieved in the presence of tris(2-furyl)phosphine (TFP) as ligand and CuI, which was assumed to facilitate the oxidative addition of the thioester. The proposed mechanism for Catellani reactions is based on the key intermediate palladacycle **69** which reacts with the electrophile to furnish the Pd^{IV} species **70**.

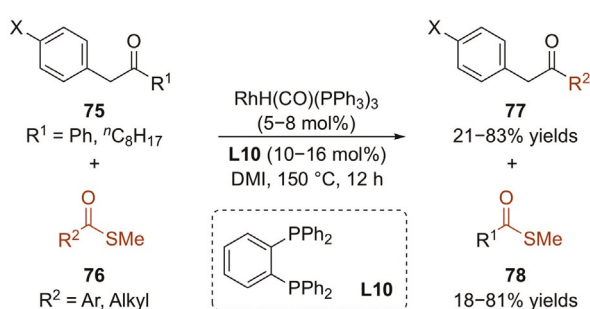
Miura, Murakami and co-workers found that a range of thioesters react with *N*-sulfonyl triazoles **71** in the presence of a rhodium(II) catalyst (Scheme 32).^[100] The resulting β -sulfanil en-



Scheme 32. Reaction of thioesters with triazines catalyzed by a Rh^{II} dimer.^[100]

amides **72** are obtained in a highly regio- and stereoselective manner. The transformation is general for a range of substituents and it was extended to thiolactones, which undergo a ring expansion reaction. According to proposed mechanism, the reaction proceeds via rhodium α -imino carbene species **73**, which then reacts with the nucleophilic sulfur atom. The resulting intermediate **74** undergoes an acyl transfer, while rhodium is released and product is formed. An intramolecular version was reported recently.^[101]

Building on their experience with acyl exchange reactions of thioesters and other carboxylic acid derivatives,^[102] Arisawa and Yamaguchi developed acyl transfer between thioesters **76** and benzylic ketones **75** (Scheme 33).^[103] The employed catalyst system consisted of RhH(CO)(PPh₃)₃ and the bidentate ligand dppBz (**L10**, = 1,2-bis(diphenylphosphino)benzene). A series of ketones with phenyl or *n*-octyl groups attached to carbonyl functionality was found to react with aromatic and aliphatic methyl thiocarboxylates in moderate to good yields. The reaction is presumably initiated by a C–C bond cleavage next to the ketone carbonyl group.

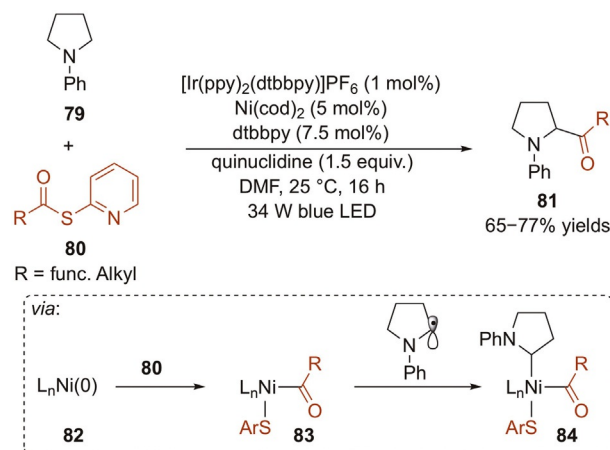


Scheme 33. Rh-catalyzed acyl transfer reaction of thioesters and ketones.^[103]

3.5. In photoredox catalysis

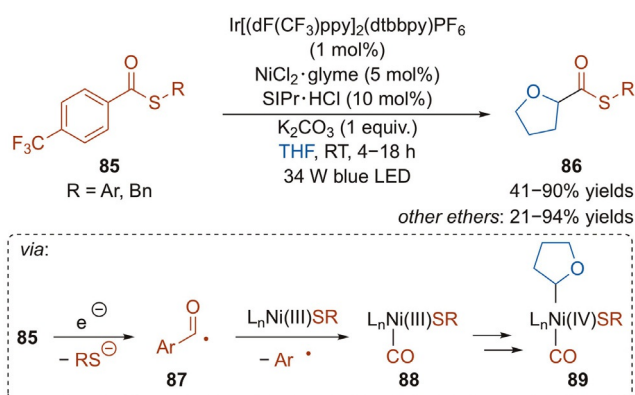
The combination of photoredox and transition metal catalysis has emerged as a powerful methodology in organic synthesis.^[104] One of its noteworthy applications is the functionalization of sp^3 C–H bonds. Thioesters possess a high potential to be applied in this type of catalysis, undergoing different activa-

tion modes. Doyle et al. disclosed a method for α -acylation of *N*-phenyl pyrrolidine (**79**) by using 2-pyridylthioesters **80** (Scheme 34).^[105] The use of thioesters constituted an atom-economic advantage over anhydrides, as it allowed for coupling of more complex acyl groups, such as a steroidal or biotine moieties. A dual catalyst system consisting of a Ni complex and Ir-based photosensitizer was employed. The suggested role of the Ni⁰ species stabilized by a bidentate N,N-ligand is the oxidative addition of the thioester and subsequent interception of the α -aminoradical. This species is generated by single-electron transfer and proton abstraction promoted by the excited Ir^{III} complex.



Scheme 34. Dual photoredox/nickel catalysis for the acylation of amines using thioesters.^[105] dtbbpy: 4,4'-di-*tert*-butyl-2,2'-dipyridyl.

In contrast, a remarkably different outcome of a similar reaction setup was achieved by Hong and Kang, who developed the first sp^3 C–H thiocarbonylation by conducting the reaction with thiobenzoates and ethers (Scheme 35).^[106] This means that the C–C(O) bond of the thioester **85** was activated instead of the weaker C–S bond. Again, an Ir-based sensitizer was used along with NiCl₂ and a carbene ligand. Various acyclic and cyclic ethers, such as THF were successfully used in reac-



Scheme 35. Thiocarbonylation of ethers by dual photoredox/nickel catalysis with electron-arm aromatic thioesters.^[106] glyme: 1,2-dimethoxyethan, SIPr·HCl: 1,3-bis(2,6-diisopropylphenyl)imidazolium chloride.

tions with thioesters decorated with aryl or benzyl groups on sulfur atom. It was essential to use electron-poor aryl groups on the acyl residue. Based on this observation, trapping experiments and calculations, author suggested a mechanistic scenario, where this time the excited Ir^{III} complex oxidizes Ni^{II} to Ni^{III} species and the resulting Ir^{II} complex donates 1 e⁻ to the thioester, which is then converted to acyl radical that undergoes a decarbonylation. Thus, the formed aryl radical abstracts a proton from the ether and the CO coordinates to the Ni^{III} complex, where remaining insertion and reductive elimination steps occur. For a detailed mechanistic discussion, we recommend the original report.

4. Conclusion

Our review highlights selected metal-catalyzed methods for the synthesis and utilization of thioesters. These compounds stand out due to their unique reactivity, which is also used in nature. The development of new methods for the synthesis of thioesters is of considerable interest for organic chemists. Especially reactions, where the carboxylic moiety is generated from well available starting materials, such as alkenes or aryl halides, are highly attractive. Indubitably, atom- and waste-economic carbonylations in the presence of thiols belong to this category. On the other hand, thioesters have huge potential as substrates in metal catalysis. In a modern context, especially the coupling reactions with or without decarbonylation are intriguing. Among these, reactions with selective but rather unreactive organometallic species are of interest if high functional group tolerance is required. Most importantly for complex molecule synthesis, S-alkyl thioesters can be carried through synthetic sequences usually unharmed in the absence of transition metal catalysts or single electron reductants. Moreover, they are generally stable to preparative chromatography on silica, bench stable and thus these functional groups compare well to other active ester functions.

However, the exact mechanistic details of many of the described transformations, especially coupling reactions, are not as well-researched as other related transformations. The literature is cautious about predicting mechanisms, as there are accumulating indications of possible one-electron redox processes operative in some catalytic systems (e.g. Marchese's Fe-catalyzed reaction). Indeed, SET steps are involved in the photoredox-catalyzed reactions. Fundamental insight into such catalytic systems, such as studies into the elementary organometallic steps^[107] or kinetic analysis, could be useful in the development of further improved variations of the known or of conceptual new transformations of thioesters.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: carbonylation • homogeneous catalysis • photocatalysis • sulfur • thioesters

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